Measurement: Sensors 27 (2023) 100755

Contents lists available at ScienceDirect

Measurement: Sensors

journal homepage: www.sciencedirect.com/journal/measurement-sensors

IOT monitoring membrane computing based on quantum inspiration to enhance security in cloud network

G. Visalaxi [a,*], A. Muthukumaravel [b]

[a] Department of Computer Science and Engineering, Bharath Institute of Higher Education and Research, Chennai, India
[b] Faculty of Arts and Science, Bharath Institute of Higher Education and Research, Chennai, India

ARTICLE INFO

Keywords:
Internet of things
Cloud computing
Quantum computing
Cryptography
Neural network
Membrane computing

ABSTRACT

Human computation is a technology inspired by nature. The majority of its computational sources are physiologically motivated computations. The existing calculation, on the other hand, uses some conventional techniques to complete the work. In the cloud, IaaS is one of the most basic services, offering a wide range of functions to a large number of customers. Modelling a number of species using P systems with different membrane structure types to predict the number of individuals is a major advantage of the proposed work. Even though it provides large number services (or) features to the user always it faces many numerous obstacles in Infrastructure service in cloud network, such as Authentication and Authorization, Data leakage and monitoring, End to End encryption, there is a risk of lack of security. Traditional encryption approaches also employ efficient methods to build cloud network security. It does, however, have some flaws in terms of IaaS. The work proposed membrane infrastructure based on the quantum inspiration. It will used to overcome such challenges. It is possible to minimize current constraints by employing this strategy. The paper proposes to employ basic distributed computational methods in an open and natural setting to provide membrane systems as a suitable framework for cloud environments for providing unique security among cloud users. They employ a framework for defending against intruder attacks. It is possible to address key cloud computing security difficulties by integrating Quantum Computing between the membrane environments with Cloud Computing technology. The proposed effectiveness and outcomes are monitored with the help of Internet of Things (IoT). This level introduces a revolutionary method to cloud security by incorporating quantum protocols into a membrane environment. The proposed SEDFA is the better method with all types of datasets and the Communication Cost by 5%, Encoding Time (ET) by 2 s, and Decoding Time (DT) by 0.5 s.

1. Introduction

Cloud computing began as an interpersonal device mean and is now broadly used for online storage and software access [1] without concern about cost of infrastructure and processing power [2]. Companies can shift their IT infrastructure to cloud and achieve quick scalability. It is essential, furthermore, for an appropriate safety strategy to be prepared for better security reasons to recognize the dangers and threats in a cloud setting [3] Organizations should have an adequate level of confidence in it into claim it. The level of consciousness was necessary to tackle cloud-related safety problems, such as virtualization, technology, protocols and perceptions of individuals. A lot of studies have been accomplished on the technical aspects of these techniques, but people's understanding and safety problems have to be achieved [4]. In cloud computing technology, traditional computing techniques have become an admirable preference. Cloud has additional access points to the cloud services, which provide elevated opportunities for various types of vulnerabilities that degrade cloud application performance. In addition to the standard risk of safety [5] of Internet-connected computer systems, cloud devices are facing accurate safety and privacy problems because of the multi-tenancy nature of the cloud. Since Cloud computing characteristics are shared by different clients, a possible malicious user can take advantage of security attacks by using information and applications of other clients for virtualization vulnerabilities. In addition, cloud users have restricted control of their information in the cloud and a cloud provider is able to manage customer information in an extreme way, including implementation data access control policies [6]. Further, information from a user can be shifted between separate cloud suppliers

* Corresponding author.
E-mail addresses: gvisalaxi@gmail.com (G. Visalaxi), dean.arts@bharathuniv.ac.in (A. Muthukumaravel).

https://doi.org/10.1016/j.measen.2023.100755
Received 11 October 2022; Received in revised form 14 March 2023; Accepted 26 April 2023
Available online 7 May 2023

in cloud computing as a result of pre-subcontract service from other providers. Moreover, a supplier of cloud information can be self-serving, untruthful and potentially malicious [7]. As a consequence, information holders hesitate to move their sensitive information to the cloud owing to privacy and data integrity issues. Therefore, the need for increased data security and secure transmission in the cloud contributes to various research methods to protect client information [8,9].

1.1. Cloud system security

Various kinds of security leaks affect any networking and computing in open web and distributed environments. This type of client server architecture and the handling of internal communication have always been important. Risk assessment is required at different cloud systems layers and components. Several internal and external attacks might happen during the processing of important data, transactions and public communication and the entire performance is under risk. Such attacks may include user level, hardware quality, and level of service or unique sharing. In order to provide information and share protection, security control procedures are required in different cloud systems layers [10–12]. Below are the concerns regarding cloud computing security features.

Hadoop, Map Reducing and Dryad platforms are extensible services provided by cloud computing. This often called as the distributed nature there will take virtualization on the substantial machines, and from that can form virtual cluster in case of large collection of virtual machines and it gives the flexible and adjustable nature to their environment, according to the changes in computation of various users, it can move up and down. Due to the involvement of malicious activities such as illegitimate users and malicious software, someone try to access sensitive data to get the benefits for others.

The main contribution of the paper is.

1. To provide a wide range of functions to a large number of customers using Human Computation technology.
2. To use P systems with different membrane structure types to model a number of species and predict the number of individuals.
3. To employ efficient methods to build cloud network security using traditional encryption approaches.
4. To overcome challenges in Infrastructure service in cloud network such as Authentication and Authorization, Data leakage and monitoring, End to End encryption by employing a quantum inspired membrane infrastructure.
5. To integrate Quantum Computing between the membrane environment with Cloud Computing technology for providing unique security among cloud users.
6. To monitor the proposed effectiveness and outcomes with the help of Internet of Things (IoT).
7. To introduce a revolutionary method to cloud security by incorporating quantum protocols into a membrane environment.
8. To improve the Communication Cost by 5%, Encoding Time (ET) by 2 s, and Decoding Time (DT) by 0.5 s in the proposed SEDFA.

The following is sectional organization of the article's body: The Basic Structural of the proposed methods are discussed in section 2 along with the Quantum Cryptography. The section 3 elaborates the proposed methodology along with its Quantum Computing. Section 4 comprehensively describes the proposed system's results and discussion with the confusion matrix. Section 6 provides the performance comparison of encoding and decoding time and communication cost. Eventually, Section 6 concludes the article.

2. Related work

Membrane Computing's original motivation is to analyse the multiple computation methods using the characteristics of living cells. This

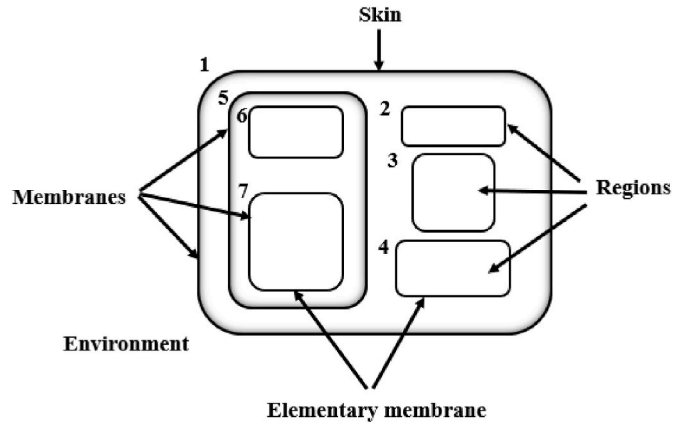

Fig. 1. Basic Structural view of P-System.

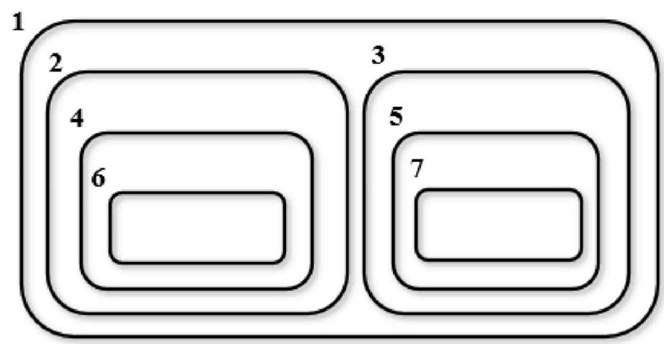

Fig. 2. Membrane structure encompasses a hierarchy of membranes.

definition was first created by Gheorghe Paun in 1998. The membrane structures are therefore referred to as P systems. In 2003, Thomson Institute described the area of membrane computing as "a rapidly growing computer science research sector." Several versions of P systems have been developed and proven computationally effective. Such complex, concurrent and decentralized computer models use time as data aid. Research in the field has grown and increased rapidly in recent decades. The new form of decentralized and parallel computing devices used in membrane system or P systems is a foundation for membrane computing [13]. The approach is based on the structure of the hierarchy: Measurable cell structures composed of the skin-enclosed cell membranes are depicted in Fig. 1.

It encompasses of many membranes structure which is arranged in some hierarchical manner with in the inner membrane structure, each of the membrane structure which is available in inner structure is called region. The outermost membrane structure is called the skin membrane [14]. It will act as a membrane wall or protection of the membrane system. The given diagram shows the various regions as illustrated in Fig. 2. The proposed membrane arrangement of Cloud service user is portrayed in Fig. 3.

The above shows the hierarchy arrangement of membrane structure (P) systems and its regions denoted as μ = {[1 [2 [4 [6]6]4]2 [3 [5 [7]7] 5]3]1}. The Polarization modes for Rectilinear and Diagonal are depicted in Figs. 4 and 5.

2.1. Quantum cryptography

Before leading the topics of Quantum let checks the existing Classical cryptography and its limitations. First there are some loopholes in the algorithm: these may be used by attackers to breach the protection of the process. The second major issue in classical cryptography is the short life expectancy algorithm. Therefore, the main size needs to be increased to

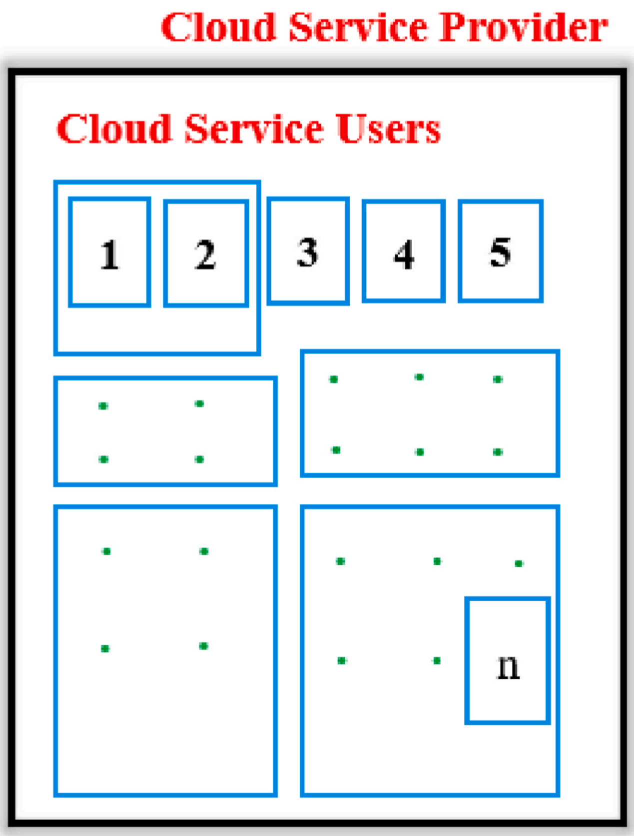

Fig. 3. Membrane arrangement of Cloud users.

$$\mu = (O, C, \mu, R_{s1}, R_{s2} \ldots R_{sn}, R_1, R_2, \ldots, R_m, io) \tag{2}$$

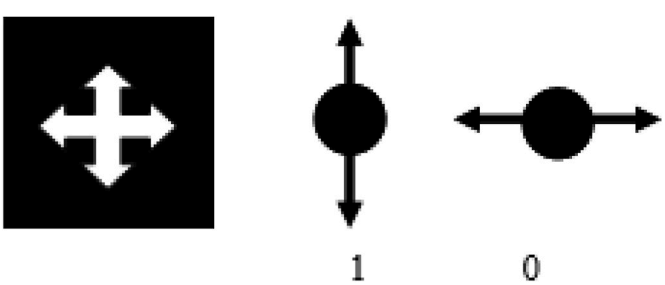

Figure: 4. Rectilinear Polarization mode.

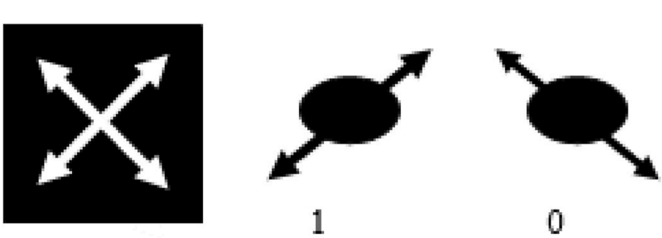

Figure: 5. Diagonal Polarization mode.

improve the life expectancy. Finally, classical authentication algorithms need more effective computation. Turn to quantum cryptography to solve this.

[15–17] explained that relay-aid scheme on BB84 protocol for the QKD process. The Q-bits are onwards to another relay node without performing any measurement the operation based on passive relays. They have analyzed the bit error rate of Quantum (QBER) by adjacent-field statistics in the relay-assisted QKD system. This type of work proposed point to point direct transmission. The results reflect the less turbulence effect. A quantum key distribution protocols (QKDPs) for security analysis. Such protocols are susceptible to dense-coding attacks. It was revealed that intruder got the session key by broadcasting entangled Q-bits as a duplicate signal to Alice. The communication between Eve and Alice has the attack process which is similar to the dense-coding communication. This did not make any errors between the information transmitted, and this was found by Alice and Bob. Finally, that the lack of certainty and the probable way to enhance these protocols have been explained in their work. The information representation is in the form of qubits in the Quantum Inspired Genetic Algorithm (QIGA). A Quantum gate acts as a divergence operator to lead people to feasible solutions. The results on this problem show the efficacy and appropriateness of an algorithm. Finally, it gives that QIGA outperforms well, compare with the traditional genetic algorithm. The merits of this algorithm are extremely stable, parallel and readily adaptable, nevertheless, the expense of computational verification. The QIGA overcomes this by increasing the population size.

The new stimulating algorithm called the Quantum Inspired Cuckoo Search Algorithm was developed by Ref. [18]. This was developed by combine features of Quantum Computing Algorithm and Cuckoo Search. It successfully solves the problems of combinational optimization using the method of Quantum computing. The proposed work combines the quantum and bio-inspired computing to produce in an effective hybrid framework, this will give the balance among the research and search capabilities. Experimental results on knapsack issues demonstrate the feasibility and the potential of the suggested system to find good performance solutions.

Drawback of the proposed method of combining quantum protocols with a membrane environment for cloud security is still in its early stages and may require further development and implementation. Furthermore, the complexity of incorporating quantum protocols into a membrane environment could be a challenge. The Research gap with further research could investigate the potential of the proposed SEDFA framework to integrate with other quantum computing protocols to enhance security in cloud computing.

3. Proposed methodology

The Unconventional Computing series like membrane and Quantum is based on the Source of inspiration from the nature. It gives always best solution rather being designed. In one of the Computing like Membrane is basically derived from a cell of our living nature. A membrane contains a living cell that will splits its internals from its exterior wall. Naturally it encompasses with artifacts such as molecules, nucleus and Golgi apparatus. It consists of charged ions which helps to move the cell for mobility. By applying the chemical reactions, the membrane allows and choose some channel for attaining the cell-to-cell signalling [19].

$$M_s = (C_r, N_e, C_{su}) \tag{1}$$

From the equation (1),

N_e -denotes to a non-empty abstract set, consisting of behavioural of user.

C_r -denotes to the allocation of cloud resource to users in cloud based environment, consisting of system structure and the resource allocated to them.

C_{su} - denotes to the Cloud Service User.

Where,

1. O is the resource available in the cloud environment,
2. $C \subset O$ is the login for user in cloud environment,
3. μ is a structure of membrane (P) system, consisting of m_i; users,

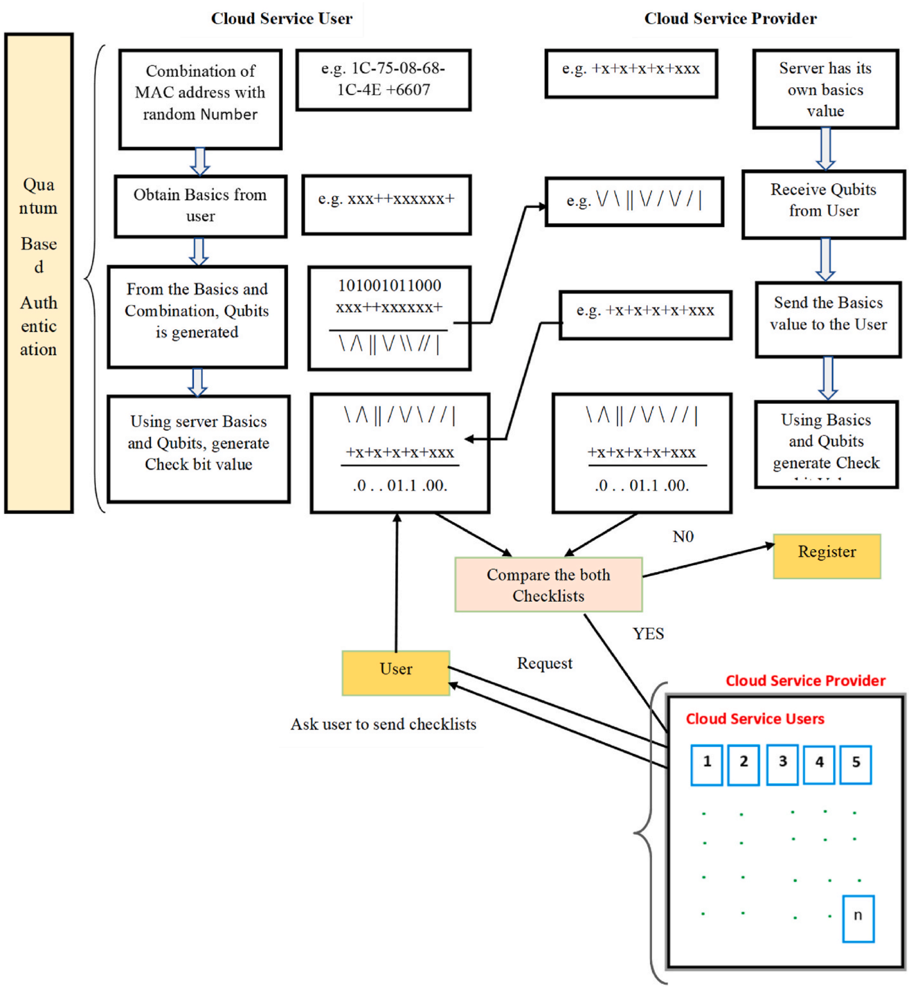

Fig. 6. Architectural diagram of Quantum inspired Membrane Computing.

labelled with 1,2 … m_{n-1};

4. R_{s1}, R_{s2} …, R_{sn} are message upon O denotes the resource for users placed in the regions 1, 2, …, S_{n-1} of the P- system,

5. R_1,R_2, …, R_m are operation associated with the user in the regions 1,2, …., m of the membrane structure,

6. io is the computation result aggregated in the environment of the system.

Every authenticated or registered user have the allocated space within the membrane-based cloud infrastructure. The allocation of space provided by CSP based on their behaviour of user or category of user. The work proposes illustrate the algorithmic dimensions of the computing power transmitted through membrane structures. The way objects are pushed in both ways is close to the form of communicating in membrane structures. The system named an anti-port membrane communication system, mainly because the membranes send and receive information. They implement many simple computer algorithms, beginning with transmitted algorithms, aggregation, ruler elections acts as an immune response mechanism for Threat attacks, and synchronization for avoiding Conflicting problem in distributed systems. The whole system works with resultant multisets or by measuring the multiplicity of Request_objects in the final stated Skin or Outer membrane.

3.1. Quantum computing

Now-a-days the communication plays a vital role in computer network. Nevertheless, secure communication between the connectivity for the computers network has now become a problem. Classical cryptography technique has been used in the present communication system of secure communication. It has more computation complexity and intruder can probable to guess what is traversing on the communication lines. This shows still lacking in traditional cryptography method. In order to minimize the probability and reduce the computational, work propose to use and implement a quantum protocol (BB84, BB92) operation for secure communication between the cloud Service user (CSU) and cloud service provider (CSP), obviously it has to be used in the cloud environment for secure communication.

In Quantum based computation, the process use Q-bits (Quantum bits) rather than Digital bits to represent the data, in addition to that work propose to utilize the secret channel for sharing the keys often called check bits in the Quantum system. It is represented mathematically by vectors such as $|0>$, $|1>$ as two states, and four bases $|H>$, $|V>$, $|R>$, $|L>$.These states are examples of what is called Qubits. By use of this, it generates the check bits in the quantum channel. someone sends Qubits |*_ that denotes either 0 or 1 [2].It cannot be figure out the value of bits it represents without known of basis it was prepared.A pair of opposite direction of Qubits which represents the basis comes from the abstract idea of qubits angular momentum often called as spins. It specify with some polarization mode, such as rectilinear and diagonal.

Above Figure: 4, it represents pair of conjugate direction and it denotes for digital representation ‘0’ for vertical and ‘1’ for horizontal (|0, V>, |1,H). And ‘0’ for 0° and ‘1’ for 90° in terms of angular momentum of spins in quantum computing.

Figure: 5 shows the diagonal polarization mode. Above it represents

Table 1
Confusion matrix.

Actual Class	Predicted Class		
		Yes	No
	Yes	TP	FN
	No	FP	TN

Table 2
Performance comparison.

Evaluation Criteria	DT	MC	Proposed QCP
Sensitivity	**95.52**	**97.01**	**99.11**
Specificity	**96.96**	**96.96**	**99.42**
Accuracy	**94.11**	**97.05**	**99.68**

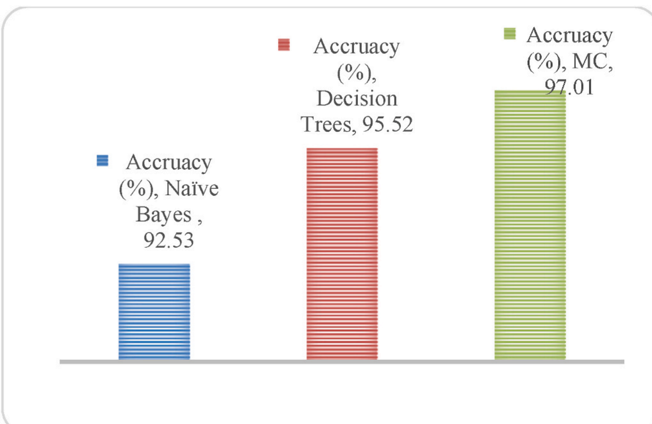

Fig. 7. Comparison of accuracy.

pair of conjugate direction and it denotes for digital representation '0' for one direction and '1' for another degree (|0,V>, |1,H). And '0' for 45° and '1' for 135° in terms of angular momentum of spins in quantum computing.

4. Performance evaluation

Membrane computing performance is evaluated by comparing the results with the results using existing systems such as Naïve Bayes and Decision Trees. In this regard, the sensitivity, specificity and accuracy are calculated in the experiment and it is considered as performance factors. Architectural diagram of Quantum inspired Membrane Computing is shown in Fig. 6. By use of confusion matrix given in Table .1 is calculated. True positive is that the number of abnormal users identified correctly as abnormal. False positive is that the number of normal users correctly identified as normal. True negative is that the number of abnormal users is not correctly identified as abnormal. False negative is that the number of normal users is not correctly identified as normal. Table-2 permits to compute the inadequate percentage of accuracy. The rows and columns in the confusion matrix signify experimental class and expected class respectively. Using the TP, TN, FP, and FN the performance factors such as accuracy, precision and recall are calculated using following equations as.

Fig. 7, it is notified that Membrane Computing attained high level of accuracy among the two types of classification. Accurate calculation of accuracy is obtained by repeatedly experimenting the MC which is a 10-fold cross validation is carried out. Finally, the accuracy obtained by Naïve Bayes, Decision Tree, and Membrane Computing is 92.53%, 95.52% and 97.01% respectively.

Again the performance of the quantum computing is evaluated and comparing the obtained results with the results using existing systems

Table 3
Performance comparison.

Methods	Malicious Detection Accuracy (%)	Delay (Sec)	Throughput (KB/Sec)
MC	97.05	26	5473
QC	99.68	15	8203

such as DNA and Quantum computing. The Same performance factors such as sensitivity, specificity and accuracy are calculated in the experiment and compared. Using the TP, TN, FP, and FN the performance factors such as accuracy, precision and recall are calculated based on the Table .2.

Table s 2 and it is explored that Quantum Computing gives highest accuracy among the both kind of classification. Finally, the accuracy obtained by DT, MC, DNA, and QC is 94.11%, 97.05%, 99.56%, and 99.68% respectively. From the percentage of accuracy, it is concluded that MC outperforms than the other methods.

The above content clearly shows the Quantum Computing cryptography method used for user identification and data security. User identification security is obtained by creating check bits based authentication passwords, and the data security is obtained by qubits. At the receiving the end qubits are extracted from the encoded information source to obtain original message. Also it replaces the computational complexity of the existing system. Thus it is identified that the proposed QC is highly suitable for user identity and data security.

5. Performance comparison

This explores the performance of the proposed work obtained at each stage. From the performance comparison it is easy to understand which algorithm outperforms than the other method. The two different security methods used in this research work are membrane computing (MC), quantum computing (QC), are experimented and the results are verified in terms of malicious detection accuracy, delay and throughput.

Table .3, it is identified and obtained that the proposed methods has outperforms individually based on their defined security tasks in cloud application. Except membrane computing all the other methods are merely equal to one another in terms of throughput, delay and accuracy. Because membrane computing has more computational complexity, it reduces the performance.

To evaluate the proposed method, three different types of datasets with size 10 KB, 100 KB, and 1 MB selected as cloud data. This approach implemented in JAVA software and the obtained results are verified. The experimental purpose JELASTIC and JAVA Net beans were used for Data deploying and testing method. Mathematical model has been proposed for implementing the proposed design to improve the security of the cloud. This method calculated the Encoding Time, Decoding Time and cost of Communication.

5.1. Encoding Time (E_nT)

Here, assume that data or Information source message M, the public key PK_i, and the set of instance of attributes considered as $input_i$. It can calculates Check bits (or) Security code CB_n with the following equation (6.1)

$$CB_n = (I, CT\ \{CT_i\} \forall i \in I \tag{3}$$

Where, $CT \rightarrow MY^s, CT_i \rightarrow A_i^s, S \rightarrow$ randomly chosen from Z_p.

5.2. Decoding time (D_nT)

Here considered the Check bits CBT encoded based on the properties set I. The secret key SK for access tree A, and found the public key as PK. The decryption procedure is described in the following equation.

Table 4
Comparison communication cost (CC) encoding time (E_nT) & decryption time (D_nT) for document, image and video database.

Learning Algorithms	10 KB			100 KB			1 MB		
	CC	E_nT	D_nT	CC	E_nT	D_nT	CC	E_nT	D_nT
Membrane computing	31	14.15	0.21	51	15	0.23	83	15.5	0.27
Quantum Computing	20	3	0.4	30	4	0.5	40	5	0.6

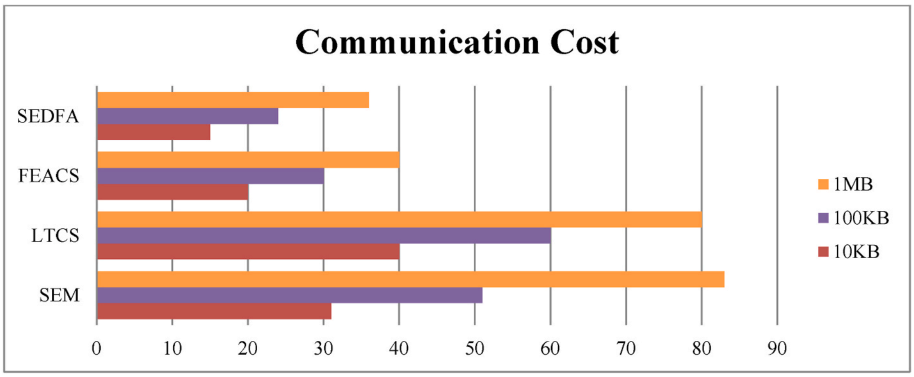

Fig. 8. Communication Cost (CC) for 10 KB, 100 KB, and 1 MB datasets.

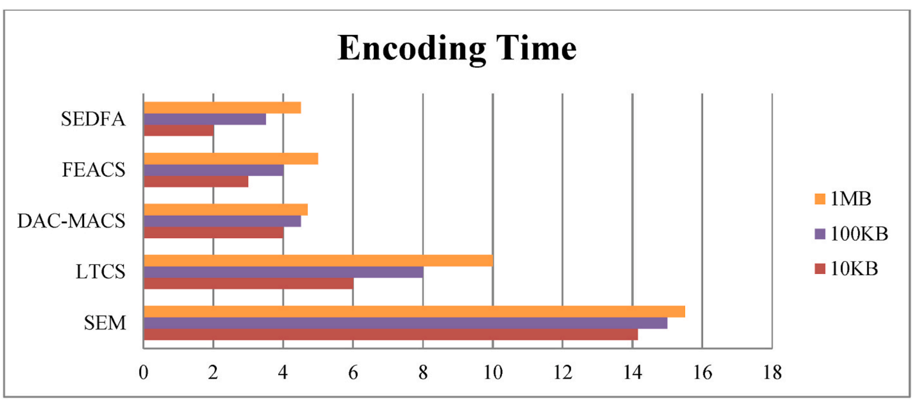

Fig. 9. Encoding Time (ET) for 10 KB, 100 KB and 1 MB datasets.

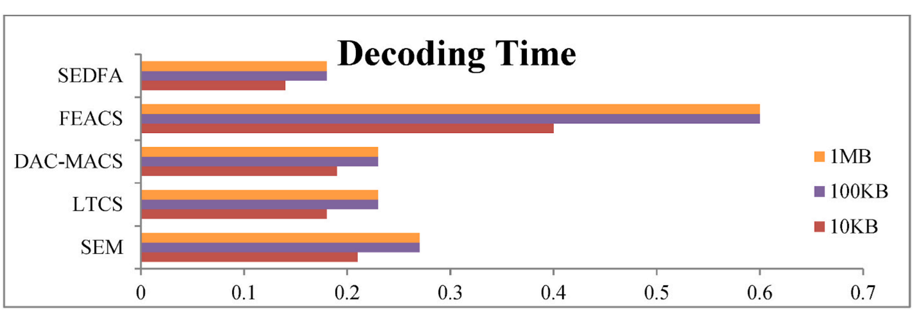

Fig. 10. Decoding Time (DT) for 10 KB, 100 KB and 1 MB datasets.

$$CT(CT_i, SK) = CT(g, g)^{pi(0)s} \tag{4}$$

Here considered CT – Cipher Text, SK - Secret Key for attribute I for leaf nodes. Then it reduces the blind factor $Y^s = CT\ (g, g)^{ys}$ and gives the message M only if *I* fulfil the criteria of A.

5.3. Communication cost (CC)

The communication cost is calculated depending upon the length of the data and the size of the index file attribute. The communication cost is c (|q|+|n|) bits given by the following equation.

$$CC = C(|q| + |n|) \tag{5}$$

Here, $|q|$ denotes the length of an element of Zp|n| denotes the length of an index.

The communication cost in percentage, the encryption time in sec, and the decryption time in the sec for the datasets 10 KB, 100 KB, and 1

MB for the existing approaches are displayed in the Table .4. The proposed approach evaluated using the communication cost (%), encryption time (in a sec) and decryption time (sec).

From the Figs. 8–10, it is analyzed that Secure and Efficient Data Forwarding Algorithm (SEDFA) performs better in all aspect when compared with all other existing methods SEM, LTCS, DACMCAS, and FEACS. It is observed that SEDFA produces the superior communication cost contrast than closest FEAFS methods. FEAFS is relatively dull for key generation and distribution process when compared to others. And also, SEDFA is better with FEAFS, but it fails to preserve encoding efficiency. Then the proposed system decryption time calculated with LTCS approaches. At last, it is concluded that the proposed SEDFA is the better method with all types of datasets and its parameters. It is analyzed that it reduces the Communication Cost by 5%, Encoding Time (ET) by 2 s, and Decoding Time (DT) by 0.5 s.

6. Conclusion

From the comparison, it is identified that this research work provides better security models for various aspects like identity based, data based, infrastructure based and data manipulation based. The four proposed methods are compared in terms of security aspects and the performance analysis is evaluated based on the metrics malicious detection, delay and throughput. From the analysis, it is identified that quantum computing outperforms than the other proposed methods. Hence, this research work is concluded as a better security method for cloud applications. The proposed scheme has used a Membrane Computing by relating its structure with cloud infrastructure to reveal the detection of abnormal users in cloud. It emphasizes reducing an abnormal user when membrane computing followed in the cloud application. Experimental results show that the proposed scheme is better than other existing schemes. Therefore, it can be concluded that increasing throughput and accuracy improves the cloud performance as a whole. Therefore, membrane computing provides better security enhancement for cloud infrastructure security than the conventional security mechanism. In future the proposed work in enhanced with defence-oriented applications.

Funding

Authors did not receive any funding.

Availability of data and material

Available on request.

Code availability

Not applicable.

CRediT authorship contribution statement

G. Visalaxi: Formal analysis, analysis and/or interpretation of data, revising the manuscript critically for important intellectual content, Approval of the version of the manuscript to be published, Conceptualization, Conception and design of study. **A. Muthukumaravel:** Formal analysis, analysis and/or interpretation of data, Writing – original draft, Drafting the manuscript, revising the manuscript critically for important intellectual content, Approval of the version of the manuscript to be published, Funding acquisition, acquisition of data.

Declaration of competing interest

Authors do not have any conflicts.

Data availability

The data that has been used is confidential.

Acknowledgment

The Author with a deep sense of gratitude would thank the supervisor for his guidance and constant support rendered during this research.

References

[1] W. Ali, J. Sang, H. Naeem, R. Naeem, A. Raza, Wireshark window authentication based packet captureing scheme to pervent DDoS related security issues in cloud network nodes, in: 2015 6th IEEE International Conference on Software Engineering and Service Science (ICSESS), IEEE, 2015, pp. 114–118.
[2] Anupriya Agarwal, Praveen Kanth, Secure data transmission using DNA encryption, Comput. Eng. Intell. Syst. 5 (7) (2017) 51–59.
[3] F. Al-Haidari, M. Sqalli, K. Salah, Evaluation of the impact of EDoS attacks against cloud computing services, Arabian J. Sci. Eng. 40 (3) (2018) 773–785.
[4] M. Ali, S.U. Khan, A.V. Vasilakos, Security in Cloud Computing: Opportunities and Challenges, vol. 305, Information sciences, 2015, pp. 357–383.
[5] M. Almorsy, J. Grundy, I. Müller, An Analysis of the Cloud Computing Security Problem, 2016 arXiv preprint arXiv:1609.01107.
[6] K. Arjunan, C.N. Modi, An enhanced intrusion detection framework for securing network layer of cloud computing, in: Proceedings of International Conference in Asia Security and Privacy, IEEE ISEA, 2017, pp. 1–10.
[7] M. Azraoui, K. Elkhiyaoui, M. Önen, K. Bernsmed, A.S. De Oliveira, J. Sendor, 'A-PPL: an Accountability Policy Language', Proceeding of the International Conference in *Data Privacy Management, Autonomous Spontaneous Security, and Security Assurance*, 2014, pp. 319–326.
[8] M. ACM Marwan, A. Kartit, H. Ouahmane, A framework to secure medical image storage in cloud computing environment, J. Electron. Commer. Org. 16 (1) (2018) 1–16.
[9] B Raj Bonny, V. Panchami, DNA based cryptography using permutation and random key generation method, International Journal of Innovative Research in Science, Engineering and Technology 3 (5) (2016) 263–267.
[10] A. Bhardwaj, G.V.B. Subrahmanyam, V. Avasthi, H. Sastry, Security algorithms for cloud computing, Proc. Comput. Sci. 85 (2016) 535–542.
[11] D. Cash, S. Jarecki, C. Jutla, H. Krawczyk, M.C. Roşu, M. Steiner, Highly-scalable searchable symmetric encryption with support for boolean queries, Proceedings of the Annual Cryptology Conference (2013) 353–373.
[12] A. Ciani, B.M. Terhal, D.P. DiVincenzo, Hamiltonian quantum computing with superconducting qubits, Quantum Sci. Technol. 4 (3) (2019), 035002.
[13] V. Chang, Y.H. Kuo, M. Ramachandran, Cloud computing adoption framework: a security framework for business clouds, Future Generat. Comput. Syst. 57 (2016) 24–41.
[14] E.F. Dumitrescu, A.J. McCaskey, G. Hagen, G.R. Jansen, T.D. Morris, T. Papenbrock, P. Lougovski, Cloud quantum computing of an atomic nucleus, Phys. Rev. Lett. 120 (21) (2018), 210501.
[15] R. Dowsley, A. Michalas, M. Nagel, N. Paladi, A survey on design and implementation of protected searchable data in the cloud, Computer Science Review 26 (2017) 17–30.
[16] M. Darwish, A. Ouda, L.F. Capretz, Formal analysis of an authentication protocol against external cloud-based denial-of-service (DoS) attack, International Journal for Information Security Research 3 (1/2) (2017) 400–407, arXiv preprint arXiv:1711.09985.
[17] M. Sohal, S. Sharma, BDNA-A DNA Inspired Symmetric Key Cryptographic Technique to Secure Cloud Computing, Journal of King Saud University-Computer and Information Sciences, 2018.
[18] S. Jarecki, A. Kiayias, H. Krawczyk, J. Xu, Highly-efficient and compassable password-protected secret sharing (or: how to protect your bitcoin wallet online), Proceedings of the IEEE European Symposium on Security and Privacy (2016) 276–291.
[19] J. Sasi Devi, R. Aarthy, M.S. DhageerBasha, A. Daniel Gnansekar, Secure data transmission in cloud computing, International Journal of Current Research and Modern Education 1 (1) (2016) 141–146.